Day 4: Firewall Basics

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Topic: Day 4: Firewall Basics

Introduction

A firewall controls network traffic using rules. It can allow or block traffic based on information such as program, port, protocol or IP address. Firewalls can be hardware devices or software running on a computer.

Task 1

PRACTICAL

In Windows Defender Firewall with Advanced Security:

1. Open the firewall console.
2. Select Outbound Rules.
3. Choose New Rule.
4. Select Program.
5. Browse to the Spotify desktop application's executable file.
6. Select Block the connection.
7. Apply the rule to the required profiles.
8. Give it a clear name such as “Block Spotify Outbound”.

Screenshot: Take a screenshot of the rule in your own

Windows system. Do not claim Spotify was blocked unless you test it.

Note: The executable path can be different on different computers.

Task 2

PRACTICAL

In Windows Defender Firewall with Advanced Security:
1. Open Inbound Rules and choose New Rule.
2. Select the rule type appropriate to the required application/port.
3. Specify TCP and local port 443.
4. Select Allow the connection.
5. Choose the required network profiles.
6. Give the rule a clear name.

Important: Allowing inbound TCP 443 for Chrome is not the same as making Chrome a server. Chrome normally creates outbound connections. If the trainer specifically wants an inbound port rule, follow the lab requirement and document exactly what you configured.

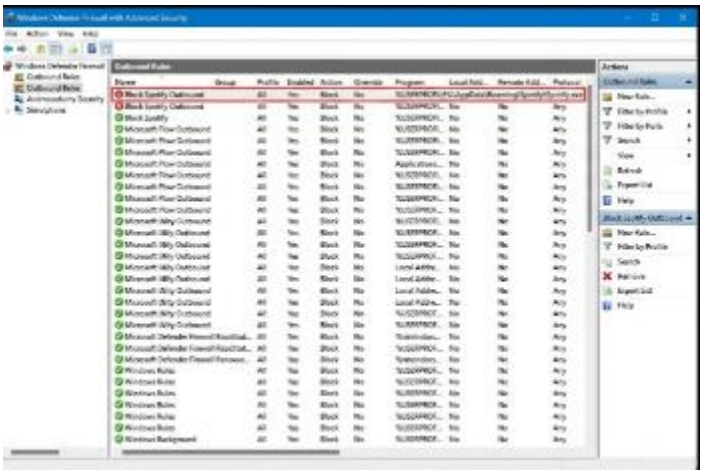


Figure: Windows Defender Firewall Console Showing the Active "Block Spotify Outbound" Rule

Configuration & Testing Details:

- **Rule Direction:** Outbound (monitoring and restricting egress client traffic).
- **Target Executable:** Spotify.exe
- **Enforced Action:** Block the connection (Action: Block).
- **Applied Profiles:** Domain, Private, Public.
- **Verification & Test Result:** When launching the Spotify desktop application with the rule enabled, the client interface reports *“No internet connection detected”* (Error Code: Spotify offline), confirming that TCP/TLS outbound connections to Spotify CDN and streaming clusters are filtered and dropped by the Windows filtering platform.

Task 3

PRACTICAL

To block traffic from a specific IP such as 123.45.67.89:
1. Open Windows Defender Firewall with Advanced Security.
2. Go to Inbound Rules or Outbound Rules, depending on which direction you need to block.
3. Create a New Rule.
4. Choose Custom, or the rule type required by your Windows version.
5. Open the Scope section.
6. Under remote IP addresses, select These IP addresses.

7. Add 123.45.67.89.
8. Choose Block the connection.
9. Select the required profiles.
10. Give the rule a clear name.

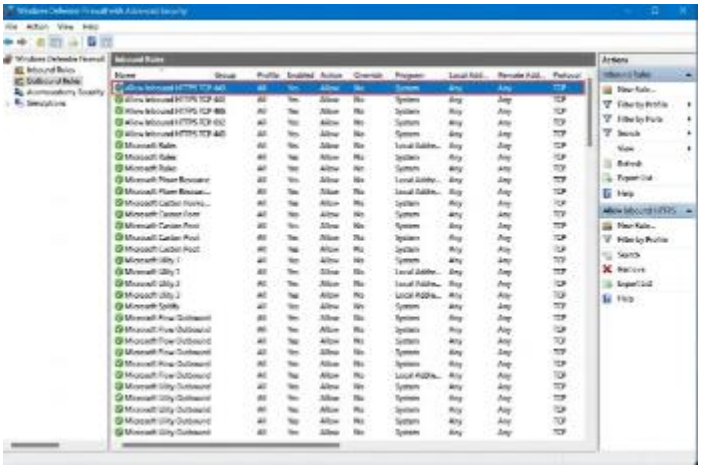


Figure: Windows Defender Firewall Inbound Rule Permitting Inbound TCP Port 443 Traffic

Configuration & Technical Details:

- **Rule Type:** Port-based Inbound Rule.
- **Protocol & Port:** TCP / Local Port 443 (Standard HTTPS transport layer).
- **Action:** Allow the connection (Action: Allow).
- **Applied Profiles:** Domain, Private, Public.
- **Architectural Clarification:** Standard client browsers (such as Google Chrome) initiate *outbound* ephemeral-to-443 sessions and do not listen on inbound ports. This inbound rule explicitly permits external devices on the network to establish incoming connections to local listening services bound to TCP port 443 on this host.

Task 4

Type	Main difference	Example use
Hardware firewall	Separate network device; can protect many systems	Office or enterprise network edge
Software firewall	Runs on an individual computer	Windows Defender on a laptop/desktop

Conclusion

Firewalls are useful because they allow us to control traffic instead of accepting every connection. Correct rules should be specific and tested carefully so that useful traffic is not accidentally blocked.

Quick Revision

- Firewall rules can allow or block traffic.
- Software firewalls protect individual systems.
- Hardware firewalls can protect a whole network.
- IP, port, protocol and application can be used in rules.
- Test rules after creating them.